FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Partograph & Labour Monitoring Tools

Clinical skill | Week 1 · Foundations, ANC & early pregnancy | Sr Kiggundu, Dr Nanzira

The partograph (and its successor, the WHO Labour Care Guide) is a one-page graphic record of maternal condition, fetal condition and labour progress that turns observations into early warnings. Plot correctly, recognise alert/action lines or LCG alert thresholds early, and act (refer, augment or deliver) before labour becomes obstructed or the baby becomes hypoxic.

Must know before the ward round

1. Modified WHO partograph - starts in active phase at 4 cm; alert line = 1 cm/hour from 4 cm; action line is 4 hours to the right of the alert line.
2. The first active-phase VE is plotted ON the alert line at that dilatation (e.g. 6 cm), with the real clock time written below.
3. Crossing the alert line at an HC III = refer; reaching the action line = full reassessment and a decision (augment, deliver, or CS).
4. Cervix plotted as X, descent (fifths palpable) as O; contractions shaded - dots under 20 s, diagonal lines 20-40 s, solid over 40 s.
5. Liquor codes - I intact, C clear, M meconium, B blood-stained, A absent. Moulding - 0, +, ++, +++.
6. Frequencies (modified partograph) - FHR, contractions and pulse every 30 min; BP every 4 h; temperature every 2 h; VE every 4 h; urine each void.
7. WHO Labour Care Guide (2020) starts at 5 cm and uses alert thresholds - e.g. no progress at 5 cm for 6 h, 6 cm 5 h, 7 cm 3 h, 8 cm 2.5 h, 9 cm 2 h; second stage 3 h (nullipara) or 2 h (multipara).
8. Normal FHR baseline 110-160 bpm; listen for a full minute immediately after a contraction; late decelerations are an alert.
9. X rising but O not falling, with moulding +++ and a big caput = CPD/obstruction - act (CS) even before the action line.
10. Oxytocin augmentation - 2.5 IU in 500 mL at 10 drops/min, up by 10 drops/min every 30 min to 60; stop if >5 contractions in 10 min or fetal distress.
11. Uganda's MoH adopted the LCG with the Essential Maternal and Newborn Clinical Care Guidelines 2022; many wards still use the modified WHO partograph - learn both.

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1. Purpose and Indications

The **partograph** (partogram) is a graphic record of labour on one page. It shows, at a glance, whether the **mother** is well, whether the **baby** is well and whether **labour is progressing**. It was designed by Philpott in Zimbabwe in the 1970s and promoted by WHO for low-resource settings, because it lets one midwife watching many women

spot trouble early and decide to **refer or act** in time.

The **WHO Labour Care Guide (LCG)**, launched in 2020, is the "next-generation partograph". It keeps the idea but updates the science (active labour from 5 cm, no rigid 1 cm/hour rule), adds **supportive care** (companion, pain relief, fluids, posture) and **shared decision-making**, and uses **alert thresholds** for each observation instead of alert and action lines.

Why it matters in Uganda: obstructed labour (leading to ruptured uterus and fistula) and intrapartum stillbirth are still common, and both are largely prevented by **monitoring labour properly and acting on what the chart shows**. A partograph that is filled in after the delivery ("retrospective partograph") helps nobody.

> CLINICAL PEARL
> The one thing never to forget: the partograph is a **decision tool, not a form**. Every time you plot a value, ask "is this normal?", and if it is not, write what you are doing about it in the plan.

Indications

- **Use for every woman in labour** in the facility (WHO), including those being augmented or induced and those on a trial of labour after caesarean.
- Start it in the **active phase only** (4 cm on the modified WHO partograph; 5 cm on the LCG).
- It is also the **referral document**: the chart goes with the woman, and the receiving facility continues it rather than starting again.
- It is a **legal record** and a key source for **maternal and perinatal death surveillance and response (MPDSR)** reviews.

> UGANDA CONTEXT
> After the **Essential Maternal and Newborn Clinical Care Guidelines (2022)**, the Ministry of Health began replacing the partograph with the **LCG** and training providers (the rollout has been gradual, with Uganda-specific modifications studied in Mbarara). Mengo and other hospitals may use either tool during the transition. The Uganda Clinical Guidelines 2023 still describe partograph use. Learn to plot and interpret **both**, and ask your ward which one is the standard.

2. Contraindications and Precautions

- **Do not use the partograph as the main tool** (or be ready to deviate immediately) when there is an indication for urgent delivery: **APH, cord prolapse, eclampsia, fetal distress, malpresentation needing CS, preterm labour below 34 weeks** (special care), or **planned caesarean**.
- **Do not start in the latent phase.** Starting in the latent phase makes normal women appear "slow" and leads to unnecessary interventions (ARM, oxytocin, caesarean).
- **Slow dilatation alone** is not a reason for oxytocin or caesarean if mother and baby are well (WHO 2018); look for the cause first.
- **CTG is not a substitute**: WHO 2018 does **not recommend admission CTG or continuous CTG for low-risk women**, because it increases caesarean and instrumental deliveries without improving newborn outcomes.
- **Oxytocin augmentation** is only for facilities with caesarean capability and one-to-one monitoring; it is contraindicated in obstruction, malpresentation, fetal distress and previous classical CS (section 8).
- **ARM** is avoided when contraindicated, for example HIV with a detectable viral load.
- A chart that is **not filled at the time** is worse than useless: it gives false reassurance. Never fill a partograph retrospectively.

3. Relevant Anatomy and Normal Values

Terms

Term	Meaning
Partograph	Graphic record of labour progress plus fetal and maternal condition
Composite WHO partograph (1994)	Had a latent phase (0-3 cm, up to 8 h) and active phase from 3 cm
Modified WHO partograph (2000)	Latent phase removed; **active phase starts at 4 cm**. Used in many Ugandan facilities and in training
Simplified partograph	Colour-coded (green/amber/red zones) versions for lower-level facilities
WHO Labour Care Guide (2020)	Starts at **5 cm**; alert thresholds for every parameter; includes supportive care and shared decision-making
Alert line	Line from 4 cm at the time of first active-phase plot, rising **1 cm per hour** to 10 cm
Action line	Parallel line **4 hours to the right** of the alert line
Intermittent auscultation (IA)	Listening to the FHR with a Pinard stethoscope or hand-held Doppler at set intervals
CTG (cardiotocography)	Continuous electronic FHR and contraction recording; not recommended for low-risk women on admission or continuously (WHO 2018)

The physiology behind the alert line

Friedman's studies (1950s) showed that once the active phase starts, the cervix of most women dilates at **at least 1 cm per hour**. Philpott used this to draw the **alert line**: a woman whose dilatation stays on or to the left of the line is progressing normally. Falling to the right means she is slower than expected, which may reflect **inadequate contractions (Powers), malposition or a large baby (Passenger), or a small pelvis (Passage)**. The **4-hour gap** before the action line gives time to refer from a health centre to a hospital (the average transfer time Philpott observed), and to try simple measures.

Modern data (Zhang and the WHO BOLD study) show many normal women dilate **more slowly than 1 cm/hour, especially before 5-6 cm**. That is why the LCG removed the fixed alert line and uses **time-at-each-centimetre thresholds** instead, and why WHO says slow dilatation **alone** is not a reason for oxytocin or caesarean if mother and baby are well.

Why each fetal parameter matters

- **FHR**: the fetus depends on placental exchange, which pauses during every contraction (the myometrium compresses the spiral arteries). A healthy fetus tolerates this; a compromised one shows **tachycardia, bradycardia or late decelerations** (the heart rate dips after the peak of a contraction and recovers late, because the fetus is becoming hypoxic during the contraction and recovering slowly).
- **Liquor**: **meconium** (especially thick) may indicate fetal hypoxia (hypoxia stimulates gut peristalsis and anal sphincter relaxation) and risks **meconium aspiration**. **Blood-stained liquor** can mean abruption or vasa praevia. **Absent liquor** after rupture means oligohydramnios, with higher risk of cord compression.
- **Moulding**: increasing moulding as labour goes on, especially +++, means the head is being squeezed through a pelvis that is too small (**CPD**).

Why the maternal parameters matter

- **Pulse** rises early in **haemorrhage, sepsis, dehydration and exhaustion**. It is the cheapest early-warning sign.

- **BP** detects pre-eclampsia (can appear first in labour) and shock.
- **Temperature** detects **chorioamnionitis** and malaria.
- **Urine**: **ketones** = dehydration/starvation (a ketotic uterus contracts poorly); **protein** = pre-eclampsia (or contamination); **low volume** = dehydration or obstruction; **blood** = bladder compression in obstructed labour.

Normal values at a glance

Parameter	Normal
FHR baseline	**110-160 bpm**, regular, no late decelerations
Contractions (adequate, active phase)	**3-5 in 10 min**, each lasting **more than 40 s**
Cervical dilatation (modified partograph)	On or left of the alert line (**at least 1 cm/h** from 4 cm)
Descent	Head progressively descending in **fifths palpable** (5/5 fully above the brim, 0/5 fully in the pelvis)
Liquor	**I** (intact) or **C** (clear)
Moulding	**0** or **+**
Maternal pulse	Below 100 (LCG alert below 60 or 120 or more)
BP	Below 140/90
Temperature	Below 37.5 degrees C
Urine	Passing regularly; no protein or acetone

4. Equipment and Preparation

Equipment

- Partograph or LCG form (one per woman), pen, watch with a second hand
- **Pinard fetal stethoscope** and/or **hand-held Doppler** (with gel, charged batteries)
- BP machine, thermometer, urine dipsticks and a measuring jug
- Sterile gloves, antiseptic, sterile pads for VEs
- A clock/timer and good light
- HemoCue (if anaemia or bleeding is suspected) and, where available, bedside ultrasound and CTG

Consent, privacy and companion

- Introduce yourself, explain each observation and **ask permission before every abdominal examination and VE**. A student must be supervised and the woman may decline a student VE.
- Close curtains; keep the woman covered except for the area examined. Offer a **female chaperone** for VEs.
- Encourage a **companion of choice**, oral fluids and mobility: these are recorded on the LCG (supportive care) and an "N" is itself an alert.

Position and exposure

- Abdominal palpation and FHR: woman semi-recumbent or in **left lateral tilt**; avoid prolonged **supine** lying (aortocaval compression; "SP" is an LCG alert).
- VE: dorsal position with knees flexed and apart, perineum exposed only for the examination.

Focused history: what the chart needs, and why

Each item changes how you plot or interpret.

Question	Why it matters to monitoring
When did regular contractions start?	Distinguishes latent vs active labour; tells you whether and where to start plotting
When did membranes rupture, and what colour was the fluid?	Write the time at the top of the chart. Over 18 h = infection risk (temperature more often, antibiotics per protocol); meconium = closer FHR monitoring
Parity and previous labours	Multiparas progress faster; a **slow multipara is obstruction until proved otherwise**
Previous caesarean	Trial of labour needs close monitoring for scar rupture (maternal pulse, scar pain, bleeding, FHR loss); any oxytocin needs senior approval
Hypertension, fever, diabetes, HIV	Adds items to monitor (BP, urine protein, temperature, glucose); affects ARM decision
Drugs already given (misoprostol for induction, oxytocin, pethidine, $MgSO_4$, antibiotics) and **when**	Tachysystole risk; neonatal respiratory depression after recent pethidine
Urge to push, bleeding, headache, fever since the last review	May bring the next VE forward
Referral partograph from a lower facility	Transfer the times and findings; do not start again "fresh"

5. Procedure Step by Step

Do it in the same order every time: **look at the woman, look at the chart, examine, plot, interpret, plan**.

5.1 Every review starts with a safety check (first 5 minutes)

Every review (and every handover) starts with a quick safety check **before** you touch the chart:

1. **Is the mother well?** Colour, distress, bleeding, fits, urge to push. Pulse now (not the last recorded value).
2. **Is the baby well?** Listen to the FHR for a full minute **straight after a contraction**, with a hand on the mother's radial pulse.
3. **Is anything running?** Oxytocin drip (rate, concentration), $MgSO_4$, IV fluids. If there are **more than 5 contractions in 10 min** or the FHR is abnormal, **stop the oxytocin now**.
4. **Look at the chart**: when was the last VE, FHR, BP, temperature and urine? Is any value already beyond the **alert line**, the **action line**, or an **LCG alert threshold** without a written plan?
5. **Call a senior at once** for: FHR below 110 or above 160 or late decelerations, cord prolapse, thick meconium, bleeding, BP 160/110 or more or fits, maternal pulse 120 or more, crossing the **action line**, or signs of obstruction.

> **RED FLAG**
> An empty partograph is itself a red flag. If the last FHR was recorded 2 hours ago, the baby has effectively not been monitored for 2 hours; listen now and document the time honestly.

5.2 Examination at each review (Hutchison order)

Follow the Hutchison sequence and examine in the **order the chart is filled**, always abdomen **before** VE.

1. **Introduce yourself, confirm identity, explain and gain consent**; offer a chaperone.

2. **Hand hygiene.**
3. **Position and exposure**: left lateral tilt or semi-recumbent; expose the abdomen only.
4. **General inspection from the end of the bed**: distress, exhaustion, colour (pallor, cyanosis), hydration (dry lips), bleeding on the pad, fits, pushing.
5. **Vital signs**: pulse (every 30 min), BP (4-hourly), temperature (2-hourly) on the modified WHO partograph.
6. **Hands, face and neck**: pallor, oedema (hands and face in pre-eclampsia), dry mucosa.
7. **Obstetric abdominal examination**:
    - **Inspection**: shape, scars (previous CS), **Bandl's ring** (obstruction), full bladder.
    - **Fundal height** and **lie**.
    - **Presentation**.
    - **Engagement / descent in fifths palpable** (plotted as **O**). Always abdominally before every VE.
    - **Contractions**: palpate for **10 full minutes**; count and time each.
    - **Auscultation of the fetal heart** for 60 s after a contraction (section 5.5).
8. **VE** (every 4 h, or sooner if indicated): dilatation (X), liquor, moulding, position, station, caput; exclude cord.
9. **Bladder and urine**: volume, protein, acetone, blood.
10. **Complete, thank the woman, cover her, hand hygiene**, then **plot, interpret and write an assessment and plan**.

Order	Examination	What to record	Key abnormal finding
1	Pulse, BP, temperature	Pulse every 30 min, BP 4-hourly, temperature 2-hourly (modified WHO partograph)	Pulse 120 or more; BP 140/90 or more; temperature 38 degrees C or more (LCG alert 37.5 or more)
2	**Descent in fifths** (abdominal)	O on the cervicograph	Head still 4/5-5/5 in active labour
3	**Contractions** by palpation for **10 full minutes**	Number of boxes shaded; dots under 20 s, diagonal 20-40 s, solid over 40 s	2 or fewer in 10 min (inadequate) or more than 5 (tachysystole)
4	**FHR** for 60 s after a contraction	Dot on the FHR graph	Below 110, above 160, late decelerations
5	**VE** (every 4 h, or sooner if indicated)	X for dilatation; liquor (I, C, M, B, A); moulding (0 to +++); position; station; caput	Cervix not advancing; moulding +++; thick meconium; cord felt; oedematous cervix
6	Bladder and urine	Volume, protein, acetone	Full bladder; ketones; protein; haematuria (obstruction)

Making each plot accurate

- **Descent**: always by **abdominal fifths** (not station alone; station can be fooled by caput).
- **Contractions**: palpate for a full 10 minutes; do not rely on the mother's report.
- **Cervix**: the same examiner at each VE reduces variability; if you are a student, have the midwife confirm.
- **Position**: identify the posterior fontanelle; an OP position explains slow progress and often a backache.

5.3 Plotting the modified WHO partograph

The chart has three parts from top to bottom: **fetal condition, labour progress, maternal condition**, plus a box for drugs and fluids.

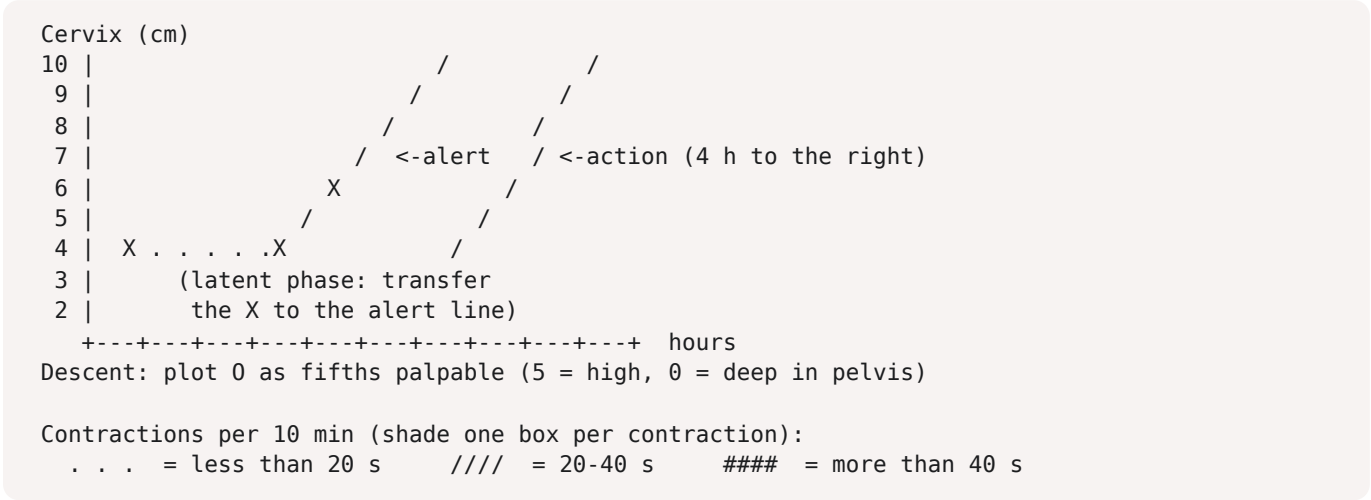
Step 1. Patient information (top). Name, age, gravida, para, hospital number, **date and time of admission**, **time of rupture of membranes**, and gestational age.

Step 2. Fetal condition.

1. **Fetal heart rate**: every **30 minutes** in the active first stage (WHO 2018: every 15-30 minutes). Listen for **a full minute immediately after a contraction**. Plot as a dot on the FHR graph (scale 80-200 bpm; the band 110-160 is often shaded) and join the dots.
2. **Liquor** at every VE (and whenever seen):
   - **I** = membranes intact
   - **C** = ruptured, clear liquor
   - **M** = meconium-stained (grade it: M1 thin, M2, M3 thick)
   - **B** = blood-stained
   - **A** = absent (ruptured, no liquor draining)
3. **Moulding** at every VE: **0** bones separate, sutures felt easily; **+** bones just touching; **++** overlapping but can be reduced by gentle pressure; **+++** overlapping and not reducible.

Step 3. Progress of labour.

1. **Cervical dilatation**: the graph has 0-10 cm on the vertical axis and hours on the horizontal axis. Plot with an **X**.
   - The **first active-phase plot (4 cm or more) goes ON the alert line**, at the matching dilatation. For example, if admitted at 6 cm, put the X at 6 cm on the alert line and write the actual clock time underneath.
   - If a woman was in the latent phase and then reaches 4 cm, the X is **transferred to the alert line** (draw a broken line to show the transfer).
   - Join successive Xs with a line.
2. **Descent of the head** by **abdominal palpation in fifths** (5/5 = fully above the brim, 0/5 = fully in the pelvis). Plot with an **O** on the same grid (the 0-5 descent scale is on the same axis). **Palpate abdominally before every VE.**
3. **Hours**: write the actual clock time under each column.
4. **Contractions**: every **30 minutes**, count the number of contractions in **10 minutes** and time the duration of each. Shade one box per contraction:
   - **Dots**: less than 20 seconds (mild)
   - **Diagonal lines**: 20-40 seconds (moderate)
   - **Solid shading**: more than 40 seconds (strong)



Step 4. Drugs and fluids.

- **Oxytocin**: record the concentration (units per litre) and the drip rate (drops/min) every 30 minutes.

- **Other drugs and IV fluids**: record every dose (antibiotics, $MgSO_4$, analgesia).

Step 5. Maternal condition.

Parameter	Frequency (modified WHO partograph)	Plot
Pulse	Every 30 min	Dot
BP	Every 4 h	Arrow (systolic top, diastolic bottom)
Temperature	Every 2 h	Number
Urine: volume, protein, acetone	Every time passed	Record

Step 6. Vaginal examinations. Every 4 hours in the active phase, or earlier if: signs of second stage, abnormal FHR, rupture of membranes (to exclude cord prolapse), bleeding, or before a decision (augmentation, referral). Each VE records cervix, liquor, moulding, station, position and caput.

5.4 Using the WHO Labour Care Guide (LCG) 2020

Structure (7 sections)

1. **Identifying information and labour characteristics at admission**: parity, labour onset (spontaneous/induced), active labour diagnosis date/time, ruptured membranes date/time, risk factors.
2. **Supportive care**: companion (Y/N), pain relief (Y/N), oral fluid (Y/N), posture (supine or mobile).
3. **Care of the baby**: baseline FHR, FHR decelerations (No, Early, Late, Variable), amniotic fluid, fetal position, caput, moulding.
4. **Care of the woman**: pulse, systolic BP, diastolic BP, temperature, urine (protein, acetone).
5. **Labour progress**: contractions per 10 min, duration of contractions, cervix (plotted as X from 5 cm), descent (plotted as O).
6. **Medication**: oxytocin (units per litre, drops/min), medicines, IV fluids.
7. **Shared decision-making**: **Assessment** and **Plan** written at each review.

How it works

- Starts in the **active first stage at 5 cm**. Observations are written in boxes (numbers or codes), not only as a graph.
- Each row has an **alert threshold** printed in the "Alert" column. If a recorded value meets the threshold, you **circle/highlight it**, and it must lead to an assessment and a documented plan (and usually a senior review).
- For cervical dilatation, instead of an alert line, there is a **time limit for each centimetre**: if the woman stays at that dilatation for the stated time or longer, it is an alert.

LCG monitoring frequencies (active first stage)

Parameter	Frequency
FHR, decelerations	Every 30 min (every 5 min in second stage)
Contractions (frequency and duration)	Every 30 min
Pulse	Hourly (more often if abnormal, per local protocol) **Verify** *confirm pulse frequency against the LCG user's manual*
BP, temperature	Every 4 h
Urine	Each void
VE (cervix, descent, liquor, position, caput, moulding)	Every 4 h
Supportive care	Every hour

> **NOTE**
> Local adaptations exist. The Uganda customisation studies simplified some sections and kept the core alert thresholds. Follow the form and schedule printed on the version your unit uses.

5.5 Intermittent auscultation of the fetal heart (the Ugandan standard)

1. Palpate to find the fetal back (anterior shoulder area); place the Pinard or Doppler there.
2. **Palpate the maternal pulse at the same time** so you do not count the mother's heart rate by mistake (especially with a Doppler, and when the mother is tachycardic).
3. Listen **during the last part of a contraction and for at least 60 seconds after it ends** (to detect late decelerations).
4. Record the **baseline rate**, rhythm, and any **decelerations** or accelerations.

Stage	Frequency
Latent phase	At least every hour (per local protocol)
Active first stage	Every 15-30 min (WHO 2018); every 30 min on the partograph
Second stage	Every 5 min, or after every contraction

6. Findings and Interpretation

Reading the lines (the geometry)

The alert line rises 1 cm per hour from the first active-phase plot, so the alert line reaches **dilatation *d* at (*d* – 4) hours after the start** (for a 4 cm start). The action line reaches the same dilatation **4 hours later**. So a woman who is at dilatation *d* at time *t*:

- is **on or left of the alert line** if she reached *d* on time (normal);
- is **between the lines** if she is up to 4 h behind;
- is **at or beyond the action line** if she is 4 h or more behind.

Position on the chart	Meaning	Action
On or left of the alert line	Normal progress	Continue monitoring and supportive care
Crossed the alert line	Slower than expected	**HC III: refer now.** Hospital/HC IV: reassess the 3 Ps, hydrate, empty bladder, mobilise, ARM if appropriate, monitor more closely
At or beyond the action line	Prolonged active phase	Senior review and a decision: augment, assisted vaginal delivery or caesarean

> **CLINICAL PEARL**
> The descent O and the dilatation X tell different stories. **X rising but O not falling** (e.g. 8 cm but head still 3/5) means the cervix is opening around a head that is not coming down: think CPD or malposition, not "good progress".

LCG alert thresholds (know these)

Section	Parameter	Alert threshold
Supportive care	Companion, pain relief, oral fluid	"N" (not provided)

Section	Parameter	Alert threshold
	Posture	"SP" (supine)
Baby	Baseline FHR	**Below 110 or 160 or more** bpm
	FHR deceleration	**L** (late)
	Amniotic fluid	**M+++** (thick meconium), **B** (blood)
	Fetal position	**Occiput posterior or transverse** (anything other than OA)
	Caput	**+++**
	Moulding	**+++**
Woman	Pulse	**Below 60 or 120 or more**
	Systolic BP	**Below 80 or 140 or more** mmHg
	Diastolic BP	**90 or more** mmHg
	Temperature	**Below 35.0 or 37.5 or more** degrees C
	Urine	**Protein ++, acetone ++**
Labour progress	Contractions per 10 min	**2 or fewer, or more than 5**
	Duration of contractions	**Below 20 or above 60 seconds**
	Cervix (time at that dilatation)	**5 cm: 6 h or more; 6 cm: 5 h; 7 cm: 3 h; 8 cm: 2.5 h; 9 cm: 2 h**
	Second stage duration	**3 h or more (nullipara); 2 h or more (multipara)**

EXAM FAVOURITE

Examiners love the LCG cervical thresholds. Memorise "**6-5-3-2.5-2**" for 5, 6, 7, 8 and 9 cm.

Fetal heart rate interpretation

Finding	Meaning / common causes
Baseline 110-160 bpm, regular, no decelerations	Normal/reassuring
Tachycardia above 160	Maternal fever/chorioamnionitis, dehydration, maternal tachycardia, drugs, early hypoxia, fetal anaemia
Bradycardia below 110 (sustained)	Cord compression/prolapse, tachysystole, abruption, uterine rupture, maternal hypotension (supine, spinal); a rate below 100 for over 3 min is an emergency
Early decelerations (dip with the contraction, mirror image)	Head compression; usually benign
Late decelerations (dip starts after the peak, recovers late)	**Uteroplacental insufficiency / hypoxia**
Variable decelerations	Cord compression; worrying if deep, prolonged or with slow recovery

CTG: when and why not routine. WHO 2018 does **not recommend admission CTG or continuous CTG for low-risk women**, because it increases caesarean and instrumental deliveries without improving newborn outcomes. CTG is useful in high-risk labours (previous CS, pre-eclampsia, oxytocin augmentation, meconium, abnormal IA) where available. Most Ugandan public units do not have it; learn IA well.

Liquor, moulding and maternal findings

Finding	Interpretation
M (especially M3 / M+++, thick)	Possible fetal hypoxia; risk of meconium aspiration; closer FHR watch, prepare for resuscitation
B	Abruption or vasa praevia until proved otherwise
A	Oligohydramnios; cord compression risk
Moulding **++**	Watch for CPD
Moulding **+++** with caput +++ and head 3/5 or more	**CPD / obstruction**
Pulse rising	Haemorrhage, sepsis, dehydration, exhaustion; in a scarred uterus, think **rupture**
Temperature 37.5 (LCG) / 38 degrees C or more	Chorioamnionitis, malaria
Urine acetone	Dehydration/ketosis (poor contractions)
Urine protein	Pre-eclampsia (or contamination)
Haematuria, low output	Obstructed labour (bladder compression)

Red flags on the chart

RED FLAG

- Dilatation crosses the **alert line** (HC III: refer) or reaches the **action line** (full assessment and decision)
- LCG alert on **cervical time thresholds** or second-stage duration
- FHR below 110 or above 160, or late decelerations
- Thick meconium (M3/M+++) or blood-stained liquor
- Moulding +++ with caput +++ and a head 3/5 or more above the brim
- More than 5 contractions in 10 minutes, or contractions lasting more than 60 seconds (especially on oxytocin)
- Maternal pulse 120 or more, temperature 37.5 or more (LCG) / 38 or more, BP 140/90 or more, protein ++, acetone ++

Bedside tests that support the chart

Tool	How	What it adds
Pinard / hand-held Doppler	60 s after a contraction, maternal pulse palpated at the same time	Baseline FHR and decelerations; avoids counting the maternal pulse
Watch with a second hand	For contractions and the FHR	Accurate counts; never estimate
Urine dipstick	Each void	Protein (pre-eclampsia), acetone (dehydration/ketosis), blood (obstruction)
Measuring jug / catheter	Measure each void	Low output in dehydration, obstruction, or pre-eclampsia
HemoCue	If pale, bleeding or known anaemia	Plan for blood before delivery
Bedside ultrasound	Position uncertain, FHR not heard, suspected malpresentation	Confirms OP/OT, breech, twins or IUFD

Tool	How	What it adds
CTG (where available)	High-risk labour or abnormal intermittent auscultation	Continuous trace; not for routine low-risk women (WHO 2018)

Laboratory and imaging when the chart raises a question

Most monitoring is at the bedside; the laboratory is used when the chart raises a question.

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC	Admission if no recent Hb; bleeding; prolonged labour; fever	Hb 11 g/dL or more; raised WBC is normal in labour, so interpret with fever and clinical signs	Anaemia → crossmatch and deliver where blood is available. HemoCue at HC III/IV; FBC at hospitals
Group and crossmatch	Crossed the action line, planned augmentation in a high-risk woman, CS likely	Blood ready	Needed before CS; HC III must refer (no blood)
Malaria RDT / smear	Temperature 37.5 degrees C or more (LCG alert) or 38 or more	Positive	Treat malaria; still look for chorioamnionitis
Urinalysis / culture	Protein ++ or fever	Proteinuria; infection	Pre-eclampsia workup or antibiotics. Culture only at regional/national referral
Platelets, LFTs, creatinine	BP 140/90 or more with protein or symptoms	Low platelets, raised ALT/AST or creatinine	Severe pre-eclampsia/HELLP → $MgSO_4$, delivery plan. Regional/national referral
Ultrasound	Slow progress with uncertain position, suspected malpresentation, FHR not found	OP/OT, brow, breech, IUFD	Explains slow progress; changes mode of delivery. Most HC IVs and hospitals, operator-dependent
CTG	Abnormal IA, oxytocin augmentation, previous CS, meconium (where available)	Normal / suspicious / pathological	Expedite delivery if pathological. Mengo, Mulago/Kawempe, some regional hospitals only
Cord blood gas	After delivery of a compromised baby (national referral only)	Acidosis	Documents intrapartum hypoxia; rarely available in Uganda

7. Clinical Acumen

Thinking like a clinician: interpreting slow progress

1. **Check the plot is correct** (started in the active phase? times right? same examiner?).
2. **Mother**: is she dehydrated, ketotic, exhausted, febrile? Is the bladder full?
3. **Powers**: count contractions; are they adequate (3-5 in 10 min, 40 s or longer)?
4. **Passenger**: position (OP/OT), deflexion, malpresentation, size; moulding and caput.
5. **Passage**: pelvic assessment; is the head still high?
6. **Baby**: FHR and liquor.

Decision matrix at the action line (or LCG cervical alert):

Findings	Likely cause	Action
Contractions inadequate, no moulding/minimal caput, head descending, **primigravida**, FHR normal	Inefficient uterine activity	Hydrate, empty bladder, ARM if membranes intact, then **oxytocin augmentation** (hospital/HC IV with CS backup), reassess in 2-4 h

Findings	Likely cause	Action
Contractions adequate, **moulding +++, big caput, head high**	CPD / obstruction	**Caesarean section**
Multipara with slow progress	Obstruction until proven otherwise	Do not give oxytocin without senior review; usually CS
FHR abnormal at any point	Fetal compromise	Intrauterine resuscitation, then expedite delivery (vacuum if fully dilated and head low; otherwise CS)
Malpresentation found (brow, face mento-posterior, transverse lie)	Mechanical	Caesarean section

If X, then think Y

- **X rising, O static** → the cervix is opening around a head that is not descending: CPD or malposition.
- **A slow multipara** → obstruction until proved otherwise; oxytocin only after senior review.
- **Doppler reading of 88 bpm** → check the maternal pulse first; you may be counting the mother.
- **Fetal tachycardia with maternal fever** → chorioamnionitis or malaria; look for the source as well as the baby.
- **Rising maternal pulse in a woman with a previous CS** → scar rupture until proved otherwise.
- **Haematuria with a high head** → obstructed labour.
- **Tachysystole on oxytocin** → stop the drip; the half-life is a few minutes.

When to stop and escalate

- **Call a senior immediately**: abnormal FHR, cord prolapse, thick meconium, bleeding, BP 160/110 or more, fits, pulse 120 or more, action line reached, signs of obstruction.
- **Do not wait for the action line** when there are signs of obstruction (moulding +++, high head, haematuria, Bandl's ring).
- **At an HC III, the alert line is the referral line.**

Worked examples

Model one-line summary from a chart: "Mrs A., P0+0 at 40 weeks, admitted at 4 cm at 08:00; now at 12:00 she is 5 cm, head 4/5, moulding ++, contractions 3 in 10 lasting 30 s, FHR 140, liquor clear; she has **crossed the alert line** and is **one hour from the action line**, mother and fetus well."

Problem list: (1) slow active phase (alert line crossed); (2) inadequate contractions; (3) moulding ++ (watch for CPD); (4) mother and fetus currently well.

Working diagnosis: "Prolonged active phase in a primigravida, probably due to inadequate uterine activity, CPD not yet excluded."

EXAM FAVOURITE

Worked example 1: crossing the lines. Admitted at **08:00 at 4 cm** (X on the alert line at 08:00). The alert line reaches 5 cm at 09:00 and 10 cm at 14:00; the action line reaches 5 cm at 13:00 and 10 cm at 18:00.

- **12:00: VE 5 cm**, head 4/5, 3 contractions in 10 min each 30 s (diagonal shading), moulding +, liquor clear, FHR 136. She is **3 h right of the alert line**, **1 h before the action line**.
- **Interpretation**: slow progress with **inadequate contractions** (not 3-5 lasting more than 40 s), no marked moulding, fetus well.
- **Plan**: at an **HC III, refer now** (alert line crossed). At hospital: rehydrate, empty the bladder, mobilise, **ARM** if membranes intact (and no contraindication), senior review; if still inadequate at the action line and no CPD → **oxytocin augmentation**; reassess in 2-4 h.

EXAM FAVOURITE

Worked example 2: obstruction before the action line. A P4 is admitted at **10:00 at 6 cm** (X on the alert line at 6 cm). At **14:00**: cervix **7 cm**, head **3/5**, **moulding +++**, **caput ++**, 4 contractions in 10 min each 50 s (solid shading), urine has blood, FHR 164.

- **Plot**: the alert line reached 7 cm at 11:00, so she is 3 h behind (between the lines).
- **Interpretation**: adequate powers, a head that is **not descending**, **non-reducible moulding**, haematuria and fetal tachycardia in a **multipara** = **obstructed labour / CPD**. You do **not** wait for the action line.
- **Plan**: no oxytocin; catheterise, IV fluids, antibiotics, crossmatch, **emergency caesarean section** (or immediate referral with an escort from an HC III).

EXAM FAVOURITE

Worked example 3: the FHR and the LCG. On the LCG, a primigravida has been at **7 cm since 09:00**; it is now 12:00 (3 h at 7 cm = **LCG cervical alert**). The FHR baseline is 145, but after each contraction it falls to 105 and returns to baseline about 40 s after the contraction has ended.

- **Interpretation**: a **late deceleration** (circle "L": an LCG alert) plus a **progress alert**.
- **Plan**: intrauterine resuscitation (stop oxytocin, left lateral, IV fluids, VE to exclude cord prolapse and assess), listen over the next 3 contractions; persistent late decelerations with a cervix at 7 cm → **caesarean section**. Write the **assessment and plan** in the shared decision-making section.

Oxytocin arithmetic you must be able to do: 2.5 IU in 500 mL = **5 mIU/mL**. With a **20-drop/mL** giving set, 10 drops/min = 0.5 mL/min = **2.5 mIU/min**. Record the IU/L and drops/min on the chart every 30 minutes.

Common student mistakes

- Plotting the **first VE of 5-6 cm at 4 cm on the time axis** instead of **on the alert line**.
- Plotting descent as **station** instead of fifths.
- Recording contractions from the mother's report instead of palpation.
- Counting the **maternal pulse** as the FHR on a Doppler.
- Listening to the FHR **between** contractions only (misses late decelerations).
- Writing values but not writing an **assessment and plan** when an alert appears.
- Filling the partograph after delivery.

8. Complications and Troubleshooting

What to do when the partograph or LCG gives an alert

At the alert line (modified WHO partograph)

1. **HC III** (no theatre): **refer** to an HC IV/hospital with the partograph, referral note, IV line and escort. Do not wait for the action line.
2. **Hospital/HC IV**: re-examine the woman (the 3 Ps), correct dehydration and bladder, encourage mobility, ARM if membranes intact and no contraindication (for example HIV with a detectable viral load), and monitor more closely.

At the action line

1. Senior review. Full assessment: maternal condition, fetal condition, contractions, pelvis, head.
2. **Decide**: oxytocin augmentation (if contractions are inadequate and there is no obstruction), **caesarean section** (obstruction, fetal distress not near delivery, malpresentation), or **assisted vaginal delivery** (fully dilated, head 0/5-1/5 and within reach).
3. Document the decision and time.

Oxytocin augmentation (hospital/HC IV only, with continuous one-to-one monitoring)

Drug	Dilution	Starting rate	Increase	Maximum	Notes
Oxytocin	**2.5 IU in 500 mL** normal saline or Ringer's lactate (5 IU/L)	**10 drops/min** (about 2.5 mIU/min with a 20-drop giving set)	By 10 drops/min every **30 min** until 3-5 contractions in 10 min, each over 40 s	60 drops/min. If still inadequate: **5 IU in 500 mL** (10 IU/L) at 30 drops/min, increasing by 10 drops/min every 30 min to 60 drops/min. If still inadequate: **multiparas and previous CS go to caesarean**; only in **primigravidae** may a 10 IU in 500 mL solution be tried (WHO MCPC regimen)	**Contraindicated** in obstruction, previous classical CS, malpresentation, fetal distress. Use with great caution in multiparas and previous lower-segment CS

DRUG DOSING

Stop the oxytocin immediately if there are **more than 5 contractions in 10 minutes**, contractions lasting **more than 60 seconds**, or **fetal distress**. Turn the woman onto her left side. If hyperstimulation persists, consider **terbutaline 250 micrograms SC** (where available). Oxytocin has a half-life of only a few minutes, so stopping the drip works quickly.

UGANDA CONTEXT

Oxytocin augmentation should only be done where caesarean section is available (HC IV and hospitals) and a health worker can stay with the woman. It is one of the commonest contributors to **ruptured uterus** in Uganda when used in multiparas or in obstructed labour.

Managing an abnormal FHR (intrauterine resuscitation)

1. **Stop oxytocin** if running.
2. **Left lateral position** (relieves aortocaval compression).
3. **IV fluids**: Ringer's lactate or normal saline (for example 500 mL-1 L) if hypotensive or dehydrated.
4. **VE**: exclude **cord prolapse**, check progress (can she deliver quickly?).
5. **Treat causes**: maternal fever (paracetamol 1 g oral, antibiotics if chorioamnionitis), hypotension, tachysystole (tocolysis if needed).

6. **Re-listen for at least 3 contractions**. If persistently abnormal: **expedite delivery**: vacuum if fully dilated and head low, otherwise caesarean section.
7. Prepare for **neonatal resuscitation** (call the person skilled in bag and mask, check equipment).

Managing prolonged second stage

- Nullipara over 3 h (or over 2 h active pushing), multipara over 2 h (or over 1 h pushing): assess for **malposition, CPD, exhaustion, inadequate contractions**.
- Options: rehydrate, empty bladder, change position, oxytocin if contractions inadequate (not in multiparas without senior review), **vacuum extraction** if the head is 0/5-1/5 and the position is known, or **caesarean** if the head is high.

Troubleshooting the tool itself

Problem	Fix
FHR not heard	Reposition over the fetal back; check Doppler battery and gel; palpate the maternal pulse; bedside ultrasound to exclude IUFD or confirm position
Doppler rate equals the maternal pulse	You are counting the mother: palpate the radial pulse at the same time and move the probe
Woman arrives from a lower facility with a partograph	Continue it; transfer times and findings; do not start "fresh"
Latent-phase woman plotted on the active graph	Re-plot: transfer the X to the alert line when she reaches 4 cm (broken line)
Different examiners give different dilatations	Same examiner where possible; students have the midwife confirm

Complications of failure to monitor or act

Missed finding	Consequence
Slow progress not acted on	Obstructed labour, **ruptured uterus**, **obstetric fistula**, maternal death
FHR abnormalities missed	Intrapartum stillbirth, **birth asphyxia/HIE**, cerebral palsy
Fever/prolonged ROM ignored	Chorioamnionitis, puerperal sepsis, neonatal sepsis
Hypertension missed	Eclampsia, stroke
Oxytocin not recorded/titrated	Tachysystole, fetal distress, uterine rupture
Over-reaction to slow early labour	Unnecessary oxytocin, ARM and caesarean

9. Documentation and Presenting Findings

- The partograph/LCG **goes with the woman** on referral.
- Write each **assessment and plan** at the time. The partograph is a legal document and a key part of **maternal and perinatal death reviews (MPDSR)**.
- On the LCG, every circled alert must have a matching **assessment** and **plan** in the shared decision-making section.

Model presentation using the partograph

"Mrs B.N., 26 years, G2P1, at 40 weeks, in labour. She was admitted at 08:00 at **5 cm**, plotted on the alert line. At 12:00 she was **9 cm**, head 1/5, moulding +, liquor clear, contractions **4 in 10 lasting 45 seconds**, FHR 130-145 throughout, pulse 80-90, BP 120/70, afebrile, passing urine without ketones. She is **on the alert line** (the alert

line runs from 5 cm at 08:00 to 9 cm at 12:00). **Assessment: normal progress with mother and fetus in good condition. Plan: continue monitoring, re-examine when she has an urge to push or if any alert appears, supportive care, prepare for delivery and AMTSL.**"

Presenting an abnormal chart (structure): identity and parity → time of admission and dilatation → current dilatation, descent, moulding, caput → contractions → FHR and liquor → maternal vitals and urine → **where she is relative to the lines** (hours behind) → assessment (the 3 Ps) → plan.

10. OSCE Checklist

Step	Done well when	Critical?
Fills patient details, time of admission and time of ROM	All identifiers, admission date/time and ROM time written at the top	No
Correctly identifies active phase (4 cm partograph / 5 cm LCG) and starts at the right point	Does not start in the latent phase	Yes
Plots first cervical dilatation **on the alert line** with X; writes clock time	X on the alert line at the matching dilatation, real time beneath	Yes
Plots descent in **fifths** with O	Uses abdominal fifths, not station	Yes
Records FHR every 30 min as dots joined by a line	Listened for 60 s after a contraction with maternal pulse checked	Yes
Records liquor (I/C/M/B/A) and moulding (0/+/++/+++) at each VE	Correct codes	No
Shades contractions correctly (dots/diagonal/solid by duration)	One box per contraction in 10 min; shading by duration	No
Records pulse, BP (arrows), temperature, urine	Correct frequencies and symbols	No
Records oxytocin (U/L and drops/min) and other drugs/fluids	Concentration and rate every 30 min	No
Identifies crossing of alert or action line and states the appropriate action	Refer at HC III; senior decision at action line	Yes
Interprets FHR abnormality and outlines intrauterine resuscitation	Stops oxytocin, left lateral, fluids, VE, re-listen, expedite	Yes
For the LCG: circles alert values and writes an assessment and plan	Every alert has a plan	Yes

11. Key Points

High-yield points

- The partograph is a **decision tool, not a form**: every abnormal value needs a written plan.
- Start in the **active phase only**: 4 cm (modified WHO partograph), 5 cm (LCG).
- The first active-phase X goes **on the alert line**; write the real clock time below.
- Alert line 1 cm/h; action line 4 h to the right. **HC III: the alert line is the referral line.**
- Descent in **fifths** (O), cervix X; **X rising but O static = CPD/malposition**.
- FHR **110-160**, listened to for **60 s after a contraction** with the maternal pulse palpated.
- LCG cervical alerts **6-5-3-2.5-2** h at 5-9 cm; second stage 3 h (nullipara) / 2 h (multipara).
- Obstruction (moulding +++, high head, haematuria) = act **before** the action line.

- Oxytocin only where CS is available; stop for >5 contractions in 10 min, >60 s or fetal distress.
- A slow multipara is obstructed until proved otherwise.
- WHO 2018: no routine admission or continuous CTG for low-risk women.

Numbers & doses

Item	Value
Start modified WHO partograph	Active phase, **cervix 4 cm or more**; first X **on the alert line**
Start LCG	**5 cm**
Alert line	**1 cm/h** from the first active-phase plot
Action line	**4 h to the right** of the alert line
FHR (first stage / second stage)	**Every 30 min** / **every 5 min** (after each contraction); normal **110-160 bpm**
Contractions and maternal pulse	**Every 30 min** (contractions counted over 10 min)
BP / temperature / VE	**4-hourly / 2-hourly / 4-hourly** (modified WHO partograph)
Contraction shading	Dots **under 20 s**, diagonal **20-40 s**, solid **over 40 s**
Adequate contractions	**3-5 in 10 min**, each over 40 s
LCG cervical alerts	5 cm **6 h or more**, 6 cm **5 h**, 7 cm **3 h**, 8 cm **2.5 h**, 9 cm **2 h**
LCG second-stage alert	**3 h or more** nullipara, **2 h or more** multipara
LCG maternal alerts	Pulse **below 60 or 120 or more**; SBP **below 80 or 140 or more**; DBP **90 or more**; temperature **below 35.0 or 37.5 or more degrees C**
LCG contraction alerts	**2 or fewer or more than 5** per 10 min; duration **below 20 or above 60 s**
Oxytocin augmentation start	**2.5 IU in 500 mL at 10 drops/min**, increase by 10 drops/min every **30 min**, max 60 drops/min
Oxytocin arithmetic	2.5 IU/500 mL = **5 mIU/mL**; 10 drops/min (20 drops/mL set) = **2.5 mIU/min**
Stop oxytocin if	**More than 5 contractions in 10 min**, contraction **over 60 s**, or fetal distress
Terbutaline (hyperstimulation)	**250 micrograms SC**
Prolonged ROM (infection watch)	**Over 18 h**
Prolonged second stage	Nullipara **over 3 h** (over 2 h pushing); multipara **over 2 h** (over 1 h pushing)

Exam traps

- **"Admitted at 6 cm: plot the X at 6 cm above the zero-hour mark."** → Wrong. The first active-phase X goes **on the alert line** at 6 cm; write the real clock time beneath.
- **"Plot descent as station."** → Wrong. Descent is plotted as **fifths palpable abdominally** (O); station can be fooled by caput.
- **"She hasn't reached the action line, so obstruction can wait."** → Wrong. Moulding +++, a high head and haematuria mean **act now**, wherever the X is.
- **"Crossed the alert line at an HC III: wait for the action line."** → Wrong. At a facility without theatre, the alert line **is** the referral line.

- **"Slow progress → oxytocin."** → Only after excluding obstruction and malpresentation, and only where CS is available; **multiparas and previous CS need senior review**.
- **"Listen to the FHR between contractions."** → Wrong. Listen during the end of a contraction and **for 60 s after it**, or you will miss late decelerations.
- **"Start the partograph at 2 cm to be safe."** → Wrong. Starting in the latent phase labels normal women as slow and drives unnecessary interventions.
- **"A Doppler reading of 88 bpm is fetal bradycardia."** → Check the **maternal pulse** first; you may be counting the mother.

Questions seniors ask

Viva questions

1. **Where do you plot the first VE of 6 cm?** On the alert line at 6 cm, and write the actual time below.
2. **What is the distance between the alert and action lines, and why?** 4 hours; originally to allow time for referral and simple interventions.
3. **What does crossing the alert line mean at an HC III?** Refer to a facility with CS capability.
4. **How do you record a contraction lasting 30 seconds?** A box with diagonal lines.
5. **What is the significance of moulding +++?** Bones overlapping and not reducible: suggests CPD.
6. **Why listen after a contraction?** To detect late decelerations (uteroplacental insufficiency).
7. **What are the LCG cervical alert thresholds?** 5 cm 6 h, 6 cm 5 h, 7 cm 3 h, 8 cm 2.5 h, 9 cm 2 h.
8. **Why did WHO move away from the 1 cm/hour rule?** It classified many normal women as slow and drove unnecessary interventions; newer evidence shows slower normal progress, especially before 5-6 cm.
9. **When is oxytocin contraindicated when the chart shows slow progress?** Obstruction/CPD, malpresentation, fetal distress, previous classical CS; high caution in grand multiparas.

On the ward round

1. **"Where is she relative to the lines?"** State the time at which the alert line reached her current dilatation, and so how many hours she is behind (0 = on the alert line; 4 or more = at or beyond the action line).
2. **"Why is she slow?"** Go through the 3 Ps plus the plot: contractions (count them), passenger (position, moulding, caput, descent), passage (pelvis), and whether the chart was started correctly.
3. **"What does moulding +++ mean?"** Overlapping skull bones that cannot be reduced: CPD until proved otherwise.
4. **"How did you record these contractions?"** Number of boxes = number in 10 min; shading by duration (dots under 20 s, diagonal 20-40 s, solid over 40 s).
5. **"What is your oxytocin rate in mIU/min?"** 2.5 IU/500 mL = 5 mIU/mL; with 20 drops/mL, 10 drops/min ≈ 2.5 mIU/min.
6. **"What would make you stop the oxytocin?"** More than 5 contractions in 10 min, a contraction lasting over 60 s, abnormal FHR, or signs of obstruction or rupture.
7. **"What is the difference between the partograph and the LCG?"** The LCG starts at 5 cm, replaces the alert/action lines with time-at-each-centimetre alerts, adds supportive care, and requires a written assessment and plan for every alert.

Mnemonics

- **Liquor codes**: "**I C M B A**" = Intact, Clear, Meconium, Blood, Absent ("I Can Monitor Babies Always").
- **Contraction shading**: "**Dots are Dainty, Lines are Light-moderate, Solid is Strong**" (under 20 s, 20-40 s, over 40 s).
- **LCG cervix thresholds**: "**6-5-3-2.5-2**" at 5-6-7-8-9 cm.
- **Plotting symbols**: "**X marks the cervix, O is the head**" (O looks like the round head going down).

- **Frequencies**: "**30-30-30, 2, 4, 4**": FHR, contractions and pulse every 30 min; temperature every 2 h; BP and VE every 4 h.
- **Slow progress**: "**3 Ps + P**": Powers, Passenger, Passage, and Plot (check the plot is right).

OSCE tips

- You may be given a **partially completed partograph** and asked to interpret it. Work top to bottom: fetal condition, progress, contractions, drugs, maternal condition. Then give a **one-line summary** and a **plan**.
- When asked "what next?", always start with "I would reassess the woman" and name the 3 Ps.
- State where she is relative to the lines **in hours**, not just "she has crossed".
- In a plotting station, say aloud why the first X goes on the alert line.

12. References

1. World Health Organization. WHO labour care guide: user's manual. Geneva: WHO; 2020. https://www.who.int/publications/i/item/9789240017566
2. World Health Organization. WHO Labour Care Guide (form). 2020. https://www.who.int/docs/default-source/reproductive-health/maternal-health/who-labour-care-guide.pdf
3. World Health Organization. WHO recommendations: intrapartum care for a positive childbirth experience. Geneva: WHO; 2018. https://www.who.int/publications/i/item/9789241550215
4. Ministry of Health Uganda. Uganda Clinical Guidelines 2023. Kampala: MoH; 2023. https://library.health.go.ug/sites/default/files/resources/Uganda%20Clinical%20Guidelines%202023.pdf
5. Ministry of Health Uganda. Essential Maternal and Newborn Clinical Care Guidelines for Uganda. 2022. https://library.health.go.ug/sexual-and-reproductive-health/essential-maternal-and-newborn-clinical-care-guidelines-uganda
6. WHO Labor Care Guide as the next generation partogram: revolutionising the quality of care during labor. 2021. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8274635/
7. Labour Care Guide implementation as a decision-making tool for monitoring labour among healthcare providers in Uganda: protocol for a mixed-methods study. 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC11029502/
8. Customization and acceptability of the WHO labor care guide to improve labor monitoring among health workers in Uganda. PLOS Global Public Health. 2024. https://journals.plos.org/globalpublichealth/article?id=10.1371%2Fjournal.pgph.0002780
9. Effectiveness of the Modified WHO Labour Care Guide to detect prolonged and obstructed labour in rural Mbarara District, southwestern Uganda. International Journal of Women's Health. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC11806744/
10. Completion of the modified World Health Organization (WHO) partograph during labour in public health institutions of Addis Ababa, Ethiopia. Reproductive Health. 2013. https://reproductive-health-journal.biomedcentral.com/articles/10.1186/1742-4755-10-23
11. FIGO consensus guidelines on intrapartum fetal monitoring: intermittent auscultation. International Journal of Gynecology & Obstetrics. 2015. https://obgyn.onlinelibrary.wiley.com/doi/10.1016/j.ijgo.2015.06.019
12. Bettercare (Perinatal Education Programme). Intrapartum Care: monitoring the condition of the mother during the first stage of labour. https://bettercare.co.za/intrapartum-care/01.html
13. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	At what cervical dilatation does the modified WHO partograph start? And the WHO Labour Care Guide?	Modified WHO partograph: **4 cm** (latent phase removed). LCG: **5 cm**.
2	How is the alert line drawn?	From **4 cm** at the time of the first active-phase plot, rising at **1 cm per hour** to 10 cm.
3	Where is the action line and why that distance?	**4 hours to the right** of (parallel to) the alert line; originally chosen to allow time to refer and to try simple interventions.
4	Symbols for plotting cervical dilatation and descent?	Cervix: **X**. Descent (fifths of head palpable abdominally): **O**.
5	Where do you plot a first active-phase VE of 6 cm?	**On the alert line** at the 6 cm level, with the actual clock time written below.
6	How are contractions recorded on the partograph?	Number in 10 minutes (one box per contraction), shaded by duration: **dots under 20 s, diagonal lines 20-40 s, solid over 40 s**.
7	Liquor codes on the partograph?	**I** intact, **C** clear, **M** meconium-stained, **B** blood-stained, **A** absent.
8	Grades of moulding?	**0** separate; **+** touching; **++** overlapping, reducible; **+++** overlapping, not reducible.
9	Monitoring frequencies on the modified WHO partograph?	FHR, contractions, pulse: **every 30 min**. Temperature: **every 2 h**. BP: **every 4 h**. VE: **every 4 h**. Urine: every void.
10	LCG cervical alert thresholds (time at each dilatation)?	**5 cm 6 h; 6 cm 5 h; 7 cm 3 h; 8 cm 2.5 h; 9 cm 2 h** (or longer).
11	LCG second-stage alert thresholds?	**3 h or more in nulliparas; 2 h or more in multiparas.**
12	LCG alert thresholds for baseline FHR and decelerations?	Baseline **below 110 or 160 and above**; **late** decelerations.
13	LCG maternal alert thresholds (pulse, BP, temperature)?	Pulse **below 60 or 120+**; SBP **below 80 or 140+**; DBP **90+**; temperature **below 35.0 or 37.5+**.
14	LCG alert thresholds for contractions?	**2 or fewer, or more than 5, per 10 min**; duration **under 20 s or over 60 s**.
15	Name the 7 sections of the LCG.	1. Identifying info/labour characteristics 2. Supportive care 3. Care of the baby 4. Care of the woman 5. Labour progress 6. Medication 7. Shared decision-making (assessment and plan).
16	How do you perform intermittent auscultation correctly?	Find the fetal back, **feel the maternal pulse simultaneously**, listen through the end of a contraction and **for at least 60 seconds after it**, record baseline, rhythm and decelerations.
17	What do late decelerations mean?	The FHR falls after the peak of a contraction and recovers late: **uteroplacental insufficiency / fetal hypoxia**.
18	Action at an HC III when the partograph crosses the alert line?	**Refer** to an HC IV/hospital (with the partograph, IV line and escort). Do not wait for the action line.
19	Starting oxytocin augmentation regimen (WHO MCPC)?	**2.5 IU in 500 mL** saline/RL at **10 drops/min**, increase by **10 drops/min every 30 min** until 3-5 contractions in 10 min each over 40 s (max 60 drops/min).

#	Question	Answer
20	Steps of intrauterine resuscitation for abnormal FHR?	**Stop oxytocin**, left lateral, IV fluids, VE to exclude cord prolapse and assess progress, treat fever/hypotension/tachysystole, re-listen; if persistent, **expedite delivery**.
21	Does WHO recommend admission or continuous CTG for low-risk women?	**No.** It increases caesarean and instrumental births without improving newborn outcomes. Use intermittent auscultation.
22	Admitted at 4 cm at 08:00. At what times does the alert line, and then the action line, reach 5 cm?	Alert line: **09:00** (1 cm/h). Action line: **13:00** (4 h later). So 5 cm at 12:00 = alert line crossed, 1 h before the action line.
23	Correct order of examination at each partograph review?	Maternal **vitals** → **descent in fifths** (abdomen) → **contractions** for 10 min → **FHR** for 60 s after a contraction → **VE** (every 4 h) → bladder/urine. Abdomen always **before** VE.
24	Cervix going from 6 to 8 cm but head still 3/5 with moulding +++. Interpretation?	X rising but O not falling: the cervix is opening around a head that is **not descending**. Think **CPD / malposition**; senior review, no oxytocin, likely CS.
25	Oxytocin 2.5 IU in 500 mL at 10 drops/min with a 20-drop/mL set: dose in mIU/min?	5 mIU/mL × 0.5 mL/min = **2.5 mIU/min**.
26	On the LCG a woman has been at 7 cm for 3 h, and the FHR drops after each contraction and recovers late. What do you do?	Two alerts: **cervical (7 cm ≥3 h)** and **late decelerations**. Intrauterine resuscitation (stop oxytocin, left lateral, IV fluids, VE for cord prolapse), senior review; persistent late decelerations at 7 cm → **caesarean section**. Write an assessment and plan.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A primigravida is admitted to Mengo labour ward at 10:00. VE shows 6 cm dilatation, 80% effaced; the head is 3/5 palpable. The ward uses the modified WHO partograph. Where should the first cervical plot be made?

- A At 6 cm on the alert line, with 10:00 written below
- B At 4 cm at 10:00, with the alert line drawn from that point
- C At 6 cm at time zero on the left margin, not on any line
- D At 6 cm on the action line
- E Do not plot until the next VE confirms progress

2. On a partograph, contractions in the last 10 minutes are recorded as 3 boxes filled with diagonal lines. What does this mean?

- A 3 contractions lasting less than 20 seconds each
- B 3 contractions per hour of moderate strength
- C 3 contractions lasting more than 40 seconds each
- D 3 contractions lasting 20-40 seconds each
- E Tachysystole

3. At an HC III, a G2P1 was plotted on the alert line at 4 cm at 08:00. At 12:00 she is 5 cm, head 3/5, contractions 2 in 10 min lasting 20-30 s, FHR 136, maternal observations normal. The nearest HC IV is 90 minutes away by road. What is the MOST appropriate next step?

- A Continue observation and reassess at 16:00
- B Refer to the HC IV now with the partograph, IV line and escort
- C Start oxytocin augmentation at the HC III
- D Perform ARM and reassess in 4 hours at the HC III
- E Encourage pushing

4. In a hospital, a primigravida reaches the action line at 7 cm. Contractions are 2 in 10 minutes lasting 25 seconds. Membranes are intact, the head is 2/5, moulding 0, no caput, FHR 140. What is the MOST appropriate management?

- A Immediate caesarean section for failure to progress
- B Give pethidine and wait
- C Vacuum delivery
- D Discharge home and review tomorrow
- E ARM followed by oxytocin augmentation with close monitoring, and review in 2-4 hours

5. A G6P5 at 40 weeks is at 7 cm, 4 hours after being plotted at 7 cm on the alert line. Contractions are 4 in 10 minutes lasting 50 seconds. The head is 3/5, moulding +++, caput ++. FHR 150. What is the BEST management?

- A Oxytocin augmentation
- B ARM and reassess in 4 hours
- C Caesarean section after resuscitation (fluids, catheter, antibiotics)
- D Vacuum extraction
- E Fundal pressure

6. A student auscultates the FHR with a Doppler and records 118 bpm, while the woman's pulse on the same chart is 118 bpm. What is the MOST important thing to do? Advanced

A Listen again while simultaneously palpating the maternal radial pulse to confirm the source
B Record it as normal and continue
C Start oxytocin
D Diagnose fetal bradycardia and prepare for caesarean
E Give oxygen by mask and wait 30 minutes

7. A woman on oxytocin augmentation has 6 contractions in 10 minutes each lasting 70 seconds. The FHR has fallen to 95 bpm and recovers slowly after each contraction. What is the FIRST action?

A Increase the oxytocin to complete the labour quickly
B Give misoprostol
C Perform a caesarean section immediately without further steps
D Stop the oxytocin infusion and turn her to the left lateral position
E Continue and recheck in 30 minutes

8. Using the WHO Labour Care Guide, a nulliparous woman was 7 cm at 09:00 and is still 7 cm at 12:00. FHR, maternal observations and contractions are normal. What does this mean?

A Normal; the LCG allows 5 hours at 7 cm
B Alert threshold reached (3 hours or more at 7 cm): assess, document and plan
C Action line crossed; immediate caesarean section
D Second stage delay
E No significance because the LCG does not track dilatation

9. Which set of LCG findings would ALL be recorded as alerts? Advanced

A FHR 150, pulse 100, temperature 37.2
B Companion present, oral fluids given, pain relief provided
C 4 contractions in 10 min, SBP 130, clear liquor
D Moulding +, caput +, occiput anterior
E FHR 165, diastolic BP 92, supine posture

10. A partograph shows cervical dilatation at 5 cm, but no descent has been plotted over 8 hours and the column for descent is blank. The midwife says she recorded 'station 0' in the notes. What is the problem?

A No problem, station is equivalent to descent
B Descent is only plotted in second stage
C Descent should be assessed by abdominal palpation in fifths and plotted as O; station alone can be misleading due to caput
D Descent is plotted as X
E Descent only matters for breech

11. A woman at 8 cm has thick green meconium-stained liquor. FHR is 150 with no decelerations. What is the MOST appropriate plan?

A Record M, increase FHR monitoring (e.g. every 15 min), watch closely for decelerations, prepare for neonatal resuscitation
B Immediate caesarean for meconium alone
C Start oxytocin
D Ignore it because the FHR is normal
E Suction the baby's mouth when the head delivers routinely, before the shoulders

12. Which is the CORRECT technique for intermittent auscultation in the active first stage?

A Listen for 15 seconds between contractions and multiply by 4
B Listen during the contraction only
C Listen once every 4 hours at the time of VE
D Listen through the end of a contraction and for at least 1 minute after it, while palpating the maternal pulse
E Listen only when the mother reports reduced movements

13. A multipara (P3) on the LCG has been fully dilated for 2 hours and 10 minutes with the head at 1/5, OA position, good contractions and a normal FHR. What is the BEST interpretation and plan? Advanced

A Normal; continue for another hour
B Second-stage alert (2 h or more in multipara): reassess, and consider assisted vaginal (vacuum) delivery if pushing is ineffective
C Immediate caesarean section
D Start oxytocin at a high rate
E Fundal pressure

14. Why does WHO no longer recommend using the 1 cm per hour alert line alone to decide on intervention?

A Because cervical dilatation cannot be measured accurately
B Because it only applies to breech presentation
C Because labour is always faster than 1 cm/hour
D Because partographs are too expensive
E Because many women with normal outcomes dilate more slowly than 1 cm/hour, especially before 5 cm, so the rule leads to unnecessary interventions

15. A woman with a previous caesarean section is on a trial of labour. At 6 cm, her pulse rises from 84 to 124 bpm, she complains of constant lower abdominal pain between contractions, there is fresh vaginal bleeding and the FHR drops to 90 bpm. What is the MOST appropriate action?

A Increase fluids and review in 1 hour
B Give pethidine for pain
C Suspect uterine rupture: call for help, resuscitate, and perform emergency laparotomy/caesarean
D Start oxytocin because contractions have weakened
E Perform ARM

Answers & explanations

> **1. A** · At 6 cm on the alert line, with 10:00 written below
>
> In the active phase, the first dilatation is plotted **on the alert line** at the corresponding level (6 cm) and the actual time is written below. Plotting at 4 cm (B) misrepresents her dilatation. Plotting at the left margin (C) is only for the old composite partograph's latent phase. The action line (D) is 4 h to the right of the alert line and would falsely suggest delay. Waiting (E) loses the baseline.

> **2. D** · 3 contractions lasting 20-40 seconds each
>
> Shading reflects **duration**: dots under 20 s, **diagonal lines 20-40 s**, solid over 40 s. The number of boxes is contractions **per 10 minutes**, not per hour (B). Under 20 s would be dots (A); over 40 s solid (C). Tachysystole (E) is more than 5 contractions in 10 minutes.

3. B · Refer to the HC IV now with the partograph, IV line and escort

She has crossed the **alert line** (expected 8 cm at 12:00). At a facility without caesarean capability, crossing the alert line is the trigger to **refer**, since the 4-hour window exists for transfer. Waiting (A, D) risks reaching obstruction far from theatre. Oxytocin (C) should be started only where CS is available and continuous monitoring is possible. Pushing (E) before full dilatation is wrong.

4. E · ARM followed by oxytocin augmentation with close monitoring, and review in 2-4 hours

The picture is **inadequate contractions** (hypotonic uterine activity) in a primigravida with a descending head and no signs of CPD (no moulding/caput). Appropriate: rehydrate, empty bladder, **ARM and oxytocin augmentation** with close monitoring. CS (A) is premature without a trial of augmentation. Vacuum (C) needs full dilatation. Discharge (D) is dangerous in active labour. Pethidine (B) does not fix inadequate contractions.

5. C · Caesarean section after resuscitation (fluids, catheter, antibiotics)

No progress despite **adequate contractions**, with moulding +++, caput and a high head in a grand multipara = **CPD/obstructed labour**: resuscitate and deliver by **caesarean section**. Oxytocin (A) risks **uterine rupture**. ARM and waiting (B) delays delivery. Vacuum (D) requires full dilatation and a low head. Fundal pressure (E) is dangerous and not recommended.

6. A · Listen again while simultaneously palpating the maternal radial pulse to confirm the source

An FHR identical to the maternal pulse suggests you may be hearing the **maternal** heart (common with Doppler, especially with maternal tachycardia or IUFD). Always **palpate the maternal pulse simultaneously**. Accepting it (B) could miss fetal death or bradycardia. Oxytocin (C) is irrelevant. Diagnosing bradycardia (D) is premature: 118 is actually within normal range if fetal. Oxygen and waiting (E) do not address the uncertainty.

7. D · Stop the oxytocin infusion and turn her to the left lateral position

This is **tachysystole with fetal distress**. First **stop oxytocin** (half-life of a few minutes) and do **left lateral positioning**, then IV fluids, VE, and consider tocolysis (terbutaline 250 mcg SC). If the FHR does not recover, expedite delivery. Increasing oxytocin (A) or giving misoprostol (B) worsens hypoxia. Going straight to CS (C) without stopping the cause wastes the chance of recovery (and the CS is safer on a recovered baby). Waiting (E) risks fetal death or uterine rupture.

8. B · Alert threshold reached (3 hours or more at 7 cm): assess, document and plan

The LCG threshold at **7 cm is 3 hours or more**, so this is an **alert**: circle it, reassess (3 Ps, hydration, bladder), write an assessment and plan (e.g. ARM/oxytocin, senior review). The 5-hour threshold (A) applies to 6 cm. The LCG has no action line and an alert does not automatically mean CS (C). She is not in second stage (D). The LCG does track dilatation (E).

9. E · FHR 165, diastolic BP 92, supine posture

LCG alerts: **FHR 160 or more**, **DBP 90 or more** and **supine posture (SP)** are all alerts. In A all values are normal. In C 4 contractions, SBP 130 and clear liquor are normal. In D moulding and caput alerts are only at +++ and OA is normal. In B all supportive care is given (the alert is 'N').

10. C · Descent should be assessed by abdominal palpation in fifths and plotted as O; station alone can be misleading due to caput

The partograph records **descent in fifths palpable abdominally, plotted as O**. Station on VE can be falsely reassuring when a big caput forms (A is wrong). Descent is plotted throughout the active phase (B is wrong). X is for the cervix (D). Descent matters for every cephalic labour and is key in recognising CPD (E).

11. A · Record M, increase FHR monitoring (e.g. every 15 min), watch closely for decelerations, prepare for neonatal resuscitation

Thick meconium (M3 / M+++ on the LCG) is an alert: it may reflect hypoxia and risks aspiration. With a normal FHR, **continue labour with closer monitoring** and prepare for resuscitation (person skilled in bag-mask present, suction ready). CS for meconium alone (B) is not indicated. Oxytocin (C) is not indicated by meconium. Ignoring it (D) is unsafe. Routine intrapartum oropharyngeal suctioning (E) is **not recommended**; it does not prevent meconium aspiration syndrome.

12. D · Listen through the end of a contraction and for at least 1 minute after it, while palpating the maternal pulse

Proper IA means listening **for at least 60 seconds after a contraction** (to detect late decelerations) and confirming it is fetal by **palpating the maternal pulse**. 15 s between contractions (A) misses decelerations. During the contraction only (B) is often inaudible and misses recovery. Four-hourly (C) is far too infrequent (should be every 15-30 min). Waiting for symptoms (E) is not monitoring.

13. B · Second-stage alert (2 h or more in multipara): reassess, and consider assisted vaginal (vacuum) delivery if pushing is ineffective

The LCG second-stage alert is **2 hours or more in multiparas** (3 h in nulliparas). With the head at 1/5, OA and a normal FHR, she is a good candidate for **vacuum** if she cannot deliver soon. Continuing without review (A) ignores the alert. CS (C) is not first-line with a low OA head. Oxytocin in a multipara in second stage (D) needs senior review and is not the first step here. Fundal pressure (E) is not recommended.

14. E · Because many women with normal outcomes dilate more slowly than 1 cm/hour, especially before 5 cm, so the rule leads to unnecessary interventions

Contemporary data show normal labour is often **slower than 1 cm/h** before 5-6 cm, so using this rule alone labels normal women as abnormal and drives oxytocin and caesarean use. Measurement is imperfect (A) but that is not the reason. C is false. Cost (D) is not the reason. It never applied only to breech (B).

15. C · Suspect uterine rupture: call for help, resuscitate, and perform emergency laparotomy/caesarean

Maternal tachycardia, constant pain, bleeding and fetal bradycardia in a woman with a uterine scar = **uterine rupture until proven otherwise**: emergency laparotomy. Waiting (A) or analgesia (B) masks the diagnosis. Oxytocin (D) worsens rupture. ARM (E) is irrelevant and delays surgery. This shows why maternal pulse is monitored every 30 minutes on the partograph.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Plotting and interpreting a normal partograph

Mrs J.M., 25 years, G2P1, 39 weeks, admitted to Mengo at 07:00 in labour. VE at 07:00: 4 cm, head 4/5, membranes intact, moulding 0. FHR 140. 2 contractions in 10 min lasting 30 s. Pulse 82, BP 118/76, T 36.9.

1. How do you plot the 07:00 findings?

> Write 07:00 under the first column. **X at 4 cm on the alert line**, **O at 4/5** for descent. Liquor **I**, moulding **0**. FHR dot at 140. Two contraction boxes shaded with **diagonal lines** (20-40 s). Pulse dot, BP arrow, temperature, and urine results in the maternal section.

2. When will you next examine her vaginally and what will you monitor meanwhile?

> Next VE at **11:00** (4 h) unless indicated earlier. Meanwhile FHR, contractions and pulse every 30 min, temperature every 2 h, BP at 11:00, urine each void. Supportive care: companion, fluids, mobility, empty bladder.

3. At 11:00: 8 cm, head 2/5, membranes ruptured at 10:00 with clear liquor, moulding +, contractions 4 in 10 lasting 45 s, FHR 132-148. Interpret.

> X at 8 cm at 11:00 is **on the alert line** (the alert line rises from 4 cm at 07:00 to 8 cm at 11:00), which is normal progress; only a plot to the **right** of the alert line signals delay. Descent is progressing (O at 2/5). Liquor **C**, moulding **+**. Contractions solid shaded. **Normal progress, mother and baby well.**

4. What is the plan now?

> Continue monitoring; expect full dilatation in the next 1-2 h. Prepare the delivery pack, **oxytocin 10 IU** for AMTSL, newborn corner (towels, bag and mask checked). Reassess if urge to push or any alert.

5. How would you present her on the ward round?

> 'Mrs J.M., 25, G2P1 at 39 weeks, in active labour, plotted at 4 cm at 07:00, now 8 cm at 11:00 with the head 2/5, on the alert line, clear liquor, mild moulding, adequate contractions, normal FHR and maternal observations. Assessment: normal progress. Plan: continue monitoring, anticipate delivery within 2 hours, AMTSL.'

Take-away: Plot the first active-phase VE on the alert line, record every section (fetal, progress, maternal), and always end with an assessment and plan.

Case 2: Crossing the action line at an HC IV

Ms P.A., 19 years, primigravida at 40 weeks, admitted to an HC IV at 5 cm at 06:00 (plotted on the alert line). At 10:00 she was 6 cm; at 14:00 she is still 6 cm. Her partograph has now crossed the action line.

1. What do you do first?

> Call the senior clinician. **Full reassessment**: maternal condition (hydration, pulse, temperature, BP, urine ketones, bladder), fetal condition (FHR, liquor), contractions (palpate for 10 min), abdominal descent, and VE for cervix, position, caput and moulding.

2. Findings: P 104, T 37.4, urine ketones ++, bladder palpable. Contractions 2 in 10 lasting 25 s. Head 3/5, LOA, moulding 0, no caput, membranes intact. FHR 145. What is your interpretation?

> **Delay in the active first stage due to inadequate uterine activity** (hypotonic contractions) with **dehydration/ketosis** and a full bladder. No features of CPD (no moulding/caput). Fetus well.

3. What is your management?

> Empty the bladder (catheter), **IV Ringer's lactate 1 L** and encourage oral fluids, **ARM** (note liquor colour), then **oxytocin augmentation**: 2.5 IU in 500 mL at 10 drops/min, increasing by 10 drops/min every 30 min to 3-5 contractions in 10 min lasting over 40 s. One-to-one monitoring; FHR every 15-30 min; reassess in 2-4 h. Document the decision.

4. Two hours later contractions are 6 in 10 minutes lasting 70 s and FHR is 100 with late decelerations. What now?

> **Tachysystole with fetal distress**: stop oxytocin, left lateral, IV fluids, VE to exclude cord prolapse and assess progress; consider terbutaline 250 mcg SC. If FHR recovers, restart oxytocin at half the rate later with care. If not, and she is not fully dilated, **emergency caesarean section**; prepare for neonatal resuscitation.

5. What lessons does this case teach about monitoring?

> The partograph identified delay; correcting reversible causes (hydration, bladder) and cautious augmentation is appropriate in a primigravida without CPD. Oxytocin must be titrated with continuous monitoring and **stopped promptly** for tachysystole or fetal distress.

> **Take-away:** At the action line, look for the cause (3 Ps and maternal state). Augment only where there is no obstruction and CS is available, and stop oxytocin at the first sign of hyperstimulation.

Case 3: Using the Labour Care Guide: supportive care and alerts

At a regional referral hospital using the LCG, Mrs D.K., 28, G3P2 at 38 weeks, was 5 cm at 08:00. The LCG shows: companion N, oral fluid N, posture SP, FHR 150-158, pulse 92, BP 138/88. At 12:00 she is 7 cm.

1. Which items are alerts and what do you do about them?

> **Companion N, oral fluid N, posture SP (supine)** are supportive-care alerts. Invite her chosen companion, offer oral fluids/light food, encourage mobility or left lateral/upright positions. Circle them and document the plan. FHR, pulse and BP are within normal limits (BP near 140/90: repeat and check urine protein).

2. At 15:00 she is still 7 cm. Is this an alert? What next?

> Yes: **3 h at 7 cm is the LCG alert threshold**. Reassess the 3 Ps: contractions, position (OP?), caput/moulding, descent, bladder, hydration. Write an assessment and plan (e.g. ARM/oxytocin if inadequate contractions; CS if obstruction), with senior review.

3. Examination shows ROP position, moulding +, caput +, head 2/5, contractions 3 in 10 lasting 45 s, BP now 146/94, urine protein ++. What are the alerts and your plan?

> Alerts: **fetal position OP**, **SBP 140 or more, DBP 90 or more, protein ++**. This is likely **pre-eclampsia developing in labour** plus a malposition causing slow progress. Plan: senior review; check for severe features (headache, visual symptoms, epigastric pain; BP 160/110), bloods (FBC/platelets, LFTs, creatinine), BP every 15-30 min; if severe features, **MgSO4** and antihypertensive per protocol. Allow mobilisation/positions to encourage rotation; decide on augmentation vs CS with the obstetrician.

4. How would you summarise the LCG status on a ward round?

> 'Mrs D.K., 28, G3P2 at 38 weeks, in active labour on the LCG since 08:00 at 5 cm; now 7 cm for 3 hours (alert), ROP, moulding +, caput +, head 2/5, adequate contractions. New hypertension 146/94 with proteinuria ++ (alert). FHR normal. Supportive care alerts earlier have been corrected. Plan: pre-eclampsia work-up and monitoring, senior decision regarding augmentation vs caesarean.'

Take-away: The LCG adds supportive care and structured alerts. Every circled value must lead to an assessment and a written plan, and maternal conditions like pre-eclampsia can appear for the first time in labour.

References

1. WHO labour care guide: user's manual. *World Health Organization*, 2020. https://www.who.int/publications/i/item/9789240017566
2. WHO Labour Care Guide (form). *World Health Organization*, 2020. https://www.who.int/docs/default-source/reproductive-health/maternal-health/who-labour-care-guide.pdf
3. WHO recommendations: intrapartum care for a positive childbirth experience. *World Health Organization*, 2018. https://www.who.int/publications/i/item/9789241550215
4. Uganda Clinical Guidelines 2023. *Ministry of Health Uganda*, 2023. https://library.health.go.ug/sites/default/files/resources/Uganda%20Clinical%20Guidelines%202023.pdf
5. Essential Maternal and Newborn Clinical Care Guidelines for Uganda. *Ministry of Health Uganda*, 2022. https://library.health.go.ug/sexual-and-reproductive-health/essential-maternal-and-newborn-clinical-care-guidelines-uganda
6. WHO Labor Care Guide as the next generation partogram: Revolutionising the quality of care during labor. *PMC*, 2021. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8274635/
7. Labour Care Guide implementation as a decision-making tool for monitoring labour among healthcare providers in Uganda: protocol for a mixed-methods study. *PMC*, 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC11029502/
8. Customization and acceptability of the WHO labor care guide to improve labor monitoring among health workers in Uganda. *PLOS Global Public Health*, 2024. https://journals.plos.org/globalpublichealth/article?id=10.1371%2Fjournal.pgph.0002780
9. Effectiveness of the Modified WHO Labour Care Guide to Detect Prolonged and Obstructed Labour in Rural Mbarara District, Southwestern Uganda. *International Journal of Women's Health / PMC*, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC11806744/
10. Completion of the modified World Health Organization (WHO) partograph during labour in public health institutions of Addis Ababa, Ethiopia. *Reproductive Health*, 2013. https://reproductive-health-journal.biomedcentral.com/articles/10.1186/1742-4755-10-23
11. FIGO consensus guidelines on intrapartum fetal monitoring: Intermittent auscultation. *International Journal of Gynecology & Obstetrics*, 2015. https://obgyn.onlinelibrary.wiley.com/doi/10.1016/j.ijgo.2015.06.019
12. Monitoring the condition of the mother during the first stage of labour (Intrapartum Care). *Bettercare (Perinatal Education Programme)*. https://bettercare.co.za/intrapartum-care/01.html

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.